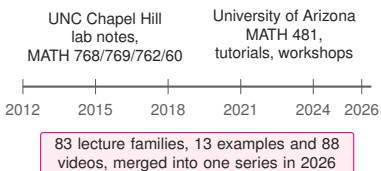
ABOUT THESE NOTES

Where the material came from, who contributed, and what was left out

Laura A. Miller

Department of Mathematics, University of Arizona

Using IB Software • Part 0 • Chapter 0.1



Claude (Anthropic) was used to organize, combine, and format more than fourteen years (2012–2026) of the instructor's tutorial notes from courses and workshops at several universities.

WHAT THESE NOTES ARE

Using IB Software is one tutorial series on the practice of immersed boundary simulation:

- IB2d: immersed boundary in 2D, in MATLAB or Python (Part II);
- IBAMR: immersed boundary in 2D and 3D on adaptive grids, in C++, including the finite element version IBFE (Parts III and IV);
- VisIt: looking at and analyzing the results (Part V);
- Planktos: agent-based models of organisms in the simulated flows (Part VI).

The companion series **Fluid Dynamics and IBM Theory** holds the equations, the discretization, and the solid mechanics behind the methods. These notes point to it and do not repeat it.

> Each chapter is one slide deck that starts with learning goals and ends with what can go wrong, a summary and references. Each worked example is a folder with tested code and a README that says what you should see.

Written for students of the University of Arizona courses and for anyone starting with these codes. You should be comfortable with a terminal and have written some code in any language.

WHERE THE MATERIAL CAME FROM

2012–2013	Miller lab IBAMR notes and “Read Me” decks at UNC Chapel Hill; MATH 769 (Spring 2013): IBAMR input files, boundary conditions, springs and targets, beams, forces, 3D, VisIt
2014	*IBAMR Tutorial: UNC Edition*; the first-simulation notes (ibamr-101); the IBFE tutorial series begins (meshes, PK1 stress, fiber tension); the gmsh geometry notes
2015	MATH 769 (Spring 2015): changing parameters, VisIt in 3D; the VisIt data-analysis guide; meshing notes (CAD to STL to mesher)
2016–2017	MATH 768 (2016) and MATH 60 (2017): IB2d and VisIt basics; gmsh 2D meshes from scratch; the IBFE series edited and extended (two meshes, 3D flow tank, advection–diffusion)
2018	MATH 769 (Spring 2018): IBAMR part 1, IBFE, visualization and data analysis, shell scripting, all recorded; the two-coral target-point example in 3D
2019	MATH 762: meshing, 3D scanning and finite elements; the 2018 VisIt lectures (2D and 3D, decks and videos) copied into a 2019 folder; the 2019 install notes
2020	The Zoom lecture series (lecture files IBAMR 1–8, IBFE 1–9, VisIt 1–5; recorded: IBAMR 1–8, IBFE 1–5, 7 and 8, VisIt 1 and 5): ConstraintIB, adding mass, Boyce’s examples, main files; the move to the University of Arizona
2021–2022	UA cluster notes (Ocelote, Puma); the summer 2021 ib2d and IBAMR tutorials and the MeshmerizeMe and Planktos decks; MATH 481 (Fall 2021: IB2d, ANSYS Fluent, CAD, scanning; Fall 2022: ANSYS Fluent and CAD); the Spring 2022 tutorials (ANSYS, CAD, scanning, IB2d and IBAMR; the decks give no course number); the IBFE summer meetings (autoibamr, Coreform Cubit)
2023–2024	MATH 481 (Fall 2023: ANSYS Fluent and CAD, no IB software); the 2023 lab tutorials (“Write your own IBM code”, a lab meeting on validating code, VisIt tutorial videos); the 2024 IB2d decks; the 2024 summer tutorials; the AIMEC–NSF Cassiopea workshop (choosing software, building geometries)
2025–2026	IBAMR 0.16.0 installed on Puma with autoibamr; IBAMR 0.19.0 released; this consolidated series, with every example rebuilt and rerun

BY THE NUMBERS

- About 150 slide decks, documents and example folders from the instructor's Google Drive, Dropbox and GitHub repositories, in several versions each.
- Grouped into 83 lecture families; the newest or most complete version of each family became the basis of a chapter.
- 21 IBAMR examples (12 IB, 6 IBFE, 3 ConstraintIB; 13 when these notes began, the others added with their chapters), each rebuilt with CMake and rerun with IBAMR 0.16.0 and 0.19.0 (IB/13 and ConstraintIB with 0.19.0 only); six IB2d folders and the Planktos scripts; the course MATLAB scripts with Python versions (Appendix A).
- 88 lecture videos (20.5 GB), listed on the following slides and not used.

> Nothing was thrown away. Material that no longer applies (retired clusters, old install recipes, old versions of the code) is in **Part X, Archive**, labelled *legacy*, and the main chapters show the current way.

The chapter decks are numbered 1–36 across Parts I–VII, with X1–X4 for the archive and A–C for the appendices. Chapter 0.2 is the map.

CONTRIBUTORS

The notes were written and taught by Laura Miller. Many people contributed material, examples and tools over the years:

- **Boyce Griffith** (UNC): IBAMR itself, its examples and documentation, and the slides from which the overview of the IB method was adapted.
- **Alex Hoover**: the IBFE tutorial series (2014–2018): PK1 stress and body forces, fiber tension, creating meshes; the tube-heart and cluster examples.
- **Nick Battista**: IB2d and its documentation, the gmsh geometry tutorial (2014), and the notes on convergence.
- **Shannon Jones**: the 3D IBAMR notes (2012) and the video how-tos on visualizing finite element meshes and recording Python in VisIt.
- **Lindsay Waldrop**: the VisIt data-analysis guide (2015) and VisIt scripts.
- **Gabrielle Hobson**: post-processing IBAMR data in VisIt on a cluster (2020).
- **Dennis Evangelista**: the first-simulation notes (2014).
- **Austin Baird**: the lab's 2012–2013 IBAMR notes.
- **Julian Wag**: gmsh 2D meshes from scratch (2016) and example READMEs.
- **Matea Santiago**: the 2023 VisIt tutorial videos and MATH 481 material.
- **Christopher Strickland**: Planktos, and the Python readers for IBAMR output.
- **D. Michael Senter**: MeshmerizeMe, meshes from images for IB2d and IBAMR.
- **Eric Tytell**: the MATLAB toolbox for reading IBAMR output.
- The students and teaching assistants of every course, whose questions and mistakes shaped the "what can go wrong" slides.

Third-party material (vendor guides, other people's decks) is credited and linked where it is used, and was rewritten rather than copied.

HOW CLAUDE WAS USED

What Claude (Anthropic) did, in 2026

- read every version of every lecture, note and example;
- merged the versions of each topic into one chapter and rewrote the text in one voice and one format;
- checked the commands and the code against IBAMR 0.16.0 and 0.19.0, rebuilt the examples with CMake, ran them, and recorded what they do;
- drafted the slides and the example READMEs, and listed what was left out.

What is the instructor's

- the material itself: the course notes, the examples, the figures and movies, and the choices of what to teach and in what order;
- the review of each chapter before it enters the main branch of the repository.

> The one-line disclosure at the bottom of every title slide says the same thing in short. If you find an error, open an issue on GitHub; the examples' READMEs say exactly what was run, so a wrong claim can be traced.

WHAT IS NOT INCLUDED

- The lecture videos. The 88 recordings from 2015 to 2024 (next three slides) are listed so that you know they exist. They are not part of these notes; new recordings will follow the new chapters.
- Research codes that are not teaching examples. Chapter 36 (Part VII) lists the examples this repository holds; teaching versions of some research models are among them (IB/10's jellyfish bell, IB/11's heart tube).
- Other people's work: student projects and theses, and third-party decks, which are cited rather than reproduced.
- Private data from the original notes: account names, passwords, e-mails and invoices were removed, and screenshots that showed them were replaced.
- Non-IB software. ANSYS, COMSOL and the CAD course are mentioned in chapter 1b for orientation and are not taught here.
- Theory. Derivations and the mathematics of the IB method moved to the companion series, *Fluid Dynamics and IBM Theory*.

LECTURE VIDEOS (1 OF 3)

The recordings below exist on the instructor's Drive (88 files, 20.5 GB in all) and are listed here only. They are not part of these notes, and new recordings will follow the new chapters. Sizes are in MB.

MATH 769 (Spring 2018): IBAMR part 1, getting started		
2-UsingSakai-and-Killdevil.mp4	2018-01-24	332
3-Part1-IBAMR-Video-ElasticBoundary.mp4	2018-01-30	130
3-Part2-IBAMR-Video-ElasticBoundary.mp4	2018-01-30	164
4-InputBoundaryConditionsPart1.mp4	2018-02-05	160
4-InputBoundaryConditionsPart2.mp4	2018-02-05	202
4-InputBoundaryConditionsPart3.mp4	2018-02-05	197
5-Video-SpringsTargetsPart1.mp4	2018-02-14	250
5-Video-SpringsTargetsPart2.mp4	2018-02-14	240
2-FirstIBAMRSimlation.mov	2018-02-23	682
6-Video_IBAMRTutorial_MultipleBoundaries_ DomainSize_InputFile.mp4	2018-03-19	229
6-Video_IBAMRTutorial_MultipleBoundaries_ DomainSize_InputFilePart2.mp4	2018-03-19	172
2-Video_IBAMRTutorial_RunOnDogwood.mp4	2018-04-13	638
MATH 769 (2018): IBFE		
1-Video_IBFE_GettingStarted.mp4	2018-04-13	451
3-Video_IBFE_MakeMeshandMoveIt.mp4	2018-04-13	728
3B-Video_IBFE_RecordPythonandRunPart1.mp4	2018-04-16	614
3B-Video_IBFE_RecordPythonandRun_Part2.mp4	2018-04-16	549

MATH 769 (2018): lectures on FSI in biology		
1-Introduction2IBM-NumericalConsiderations.mp4	2018-01-24	841
2-IntroDimensionlessParameters.mp4	2018-02-11	501
MATH 769 (2018): professional development		
LiteratureSearchPart1.mp4	2018-02-11	346
LiteratureSearchPart2.mp4	2018-02-11	235
Visualization and data analysis (2018)		
1-2DVisitVisualization.mp4	2018-02-16	471
2-Video_Visualizing3Ddata.mp4	2018-02-26	258
4-Video_VisitDataAnalysis-SamplingandTimeQueries.mp4	2018-03-19	323
4-Video_VisitDataAnalysis-SamplingandTimeQueriesPart2.mp4	2018-03-19	208
5-Video_IBAMR_MATLAB_Analysis_Part1.mp4	2018-03-19	219
5-Video_IBAMR_MATLAB_Analysis_Part2.mp4	2018-03-19	370
6-Video_ResamplingAMR2UniformGrid_ Exporting2vtk_RecordingInPython.mp4	2018-03-19	523
7-Video_IBAMR_RunningManySims1A_ Intro2ShellScripting.mp4	2018-03-25	523
7-Video_IBAMR_RunningManySims1B_ Intro2ShellScripting.mp4	2018-03-25	662
7-Video_IBAMR_RunningManySimsPart2_ Intro2ShellScripting.mp4	2018-04-02	738

LECTURE VIDEOS (2 OF 3)

8-Visit from command line and Python.mp4	2018-04-02	809
2019 VisIt lectures (copies of the 2018 videos)		
Lecture 1-2DVisitVisualization.mp4	2019-07-03	471
Lecture 2-Video_Visualizing3Ddata.mp4	2019-07-03	258
2020 Zoom lectures (IBAMR, IBFE, VisIt)		
Lecture 1 - Running your first simulation.mp4	2020-05-11	93
Visualization Lecture 1 - Getting Started with VisIt. mp4	2020-05-12	125
Lecture 2 - Intro to IBM.mp4	2020-05-13	91
Lecture 3 - IBAMR input files.mp4	2020-05-14	104
Lecture 4 - Boundary conditions, domain sizes, etc.. mp4	2020-05-15	70
Lecture 5 - Moving immersed boundaries.mp4	2020-05-19	69
IBFE-Lecture1-BeamExample.mp4	2020-05-20	63
IBFE-Lecture2-MakeYourOwnMesh.mp4	2020-05-26	77
IBFE-Lectures 3 and 4 - Active stresses.mp4	2020-05-27	77
IBFE-Lecture5 - Plans for June.mov	2020-06-01	69
Lecture 6 - What do Boyce's examples do.mp4	2020-07-21	58
IBFE-Lecture7-TwoBoundaries.mp4	2020-08-10	40
IBFE-Lecture 8 - Main Files.mp4	2020-09-01	73
Lecture 7 - Constraint IB with Sound.mp4	2020-09-09	50
Lecture 7 - No Sound - Constraint IB.mp4	2020-09-09	118
Visualization Lecture 5 - Visit on the cluster.mp4	2020-09-23	81
Lecture 8 - Adding massive points.mp4	2020-11-11	42
IBFE-Lectures 3 and 4 - Active stresses.mp4	2021-06-25	77
Hamlet_Run_IBAMR.mp4	2021-08-04	9
2020, other		
Gmsh2D.mp4	2020-10-02	37
2022 CFD tutorials (ANSYS, CAD, Meshmixer; Tutorial 1 also IB2d and IBAMR)		
Tutorial1-ANSYSandIBAMR.mp4	2022-01-18	367
Tutorial3 - Meshmixer.mp4	2022-02-04	372
Tutorial4A - Cleaning Meshes.mp4	2022-02-15	372
Tutorial4B-MeshCleaningANSYSsimulation.mp4	2022-02-15	309
Tutorial2 - Autodesk Fusion.mp4	2022-02-24	343
ANSYSQuickOverviewTrachea.mp4	2022-04-21	176
2022 IBFE summer meetings		
July-21-2022-Installing-autoibamr.mp4	2022-07-21	157

LECTURE VIDEOS (3 OF 3)

June-23-2022-InitialMeetingOverview.mp4	2022-07-21	306
June-30-2022-QuickActiveTensionReview.mp4	2022-07-21	97
July-28-2022-Coreform-Tutorial.mp4	2022-07-29	345
2022 ib2d and IBAMR tutorials		
Tutorial2-Boundaries-ib2d-IBAMR.mp4	2022-01-28	240
Tutorial3-MovingBoundaries-ib2d.mp4	2022-02-04	128
Tutorial1B-RunningIBAMR.mp4	2022-02-24	161
GenerateMeshTutorialPart1.mp4	2022-04-21	182
GenerateMeshTutorialPart2.mp4	2022-04-21	267
2023 VisIt tutorial (M. Santiago)		
2D Slices.mp4	2023-07-25	61
3D flow.mp4	2023-07-25	40
Finalizing Visualizations.mp4	2023-07-25	48
Intro-Mesh-Body.mp4	2023-07-25	18
2023 "Write your own IBM code" (a simulation movie)		
fluid_movie_16.avi	2023-05-07	9
2024 summer tutorials: IBAMR		
IBAMR-BCs-July11.mp4	2024-07-11	131
IBAMR-DomainSize-July16.mp4	2024-07-16	233
IBAMR-intro-running-on-hpc-July3.mp4	2024-07-16	243
IBAMR-MovingTarget-ChangeSprings.mp4	2024-07-28	201
2024 summer tutorials: IBFE		
0-IBFE-Getting-Started.mp4	2024-08-01	121
1-IBFE-TargetPlateExample.mp4	2024-08-01	355
2-IBFE-MakeMeshMoveIt.mp4	2024-08-01	67
2024 summer tutorials: ib2d		
ib2d-intro-07-02-2024.mp4	2024-07-02	250
2015 tube-heart simulation movies (lab project; not lectures)		
tubemovie_oneactivation.mpg	2015-08-21	4
tubemovie_1hz_activation.mpg	2015-08-24	7
tubemovie_1hz_activation_altview.mpg	2015-08-24	3
tubemovie_oneactivation_altview_mu1e5_T2e5.mpg	2015-08-27	3
tubemovie_oneactivation_mu1e5_T2e5.mpg	2015-08-27	7
tubemovie_peristalsis_altview_mu1e6_T1e6.mpg	2015-10-19	3
tubemovie_peristalsis_mu1e6_T1e6.mpg	2015-10-19	6

LICENSE, CITATION AND WHERE THINGS ARE

License

- Text, slides and figures: Creative Commons Attribution 4.0 (CC BY 4.0).
- Code in `examples/`: BSD 3-Clause.
- Third-party items keep their own licenses, named where they appear.

To cite these notes

Miller, L. A. (2026). *Using IB Software: tutorials for IB2d, IBAMR, VisIt and Planktos.* https://github.com/fairyflies9/Using-IB-Software

Where things are

- These notes: https://github.com/fairyflies9/Using-IB-Software
- EquiPhysics, the author's nonprofit, which brings horse lovers to mathematics and shares these notes: https://equiphysics.org
- The theory series: https://github.com/fairyflies9/Fluid-Dynamics-and-IBM-Theory
- IBAMR: https://ibamr.github.io
- IB2d: https://github.com/nickabattista/IB2d
- VisIt: https://visit-dav.github.io/visit-website/
- Planktos: https://github.com/mountaindust/Planktos

Next, chapter 0.2: the map of the chapters, the software versions used, and where the examples live.